IndianOil
A Maharatna Company

Indian Oil Corporation Limited
Pipelines Division, WRPL

NOTE

Ref: WRPL/RAJ/MNT/ Date: 29.01.2014

Sub: Failure of newly commissioned Booster Pump at Vadinar in WRPL.

1) This has been reference to preliminary report on failure of newly commissioned Booster Pump at Vadinar received from WRPL Vadinar vide mail dated 29.01.2014. Copy enclosed. Report has been reviewed and observations are as below.

2) Five nos of new BPCL make booster pumps were installed at Vadinar as part of additional tank farm facility commissioned in the month of April 2013. BP1, BP2 and BP3 are identical booster units and having the rated flow rate of 2000 kl/hr and 02 booster pumps BP4 & BP5 are having flow rate of 1000 kl/hr.

3) **Three boosters of higher capacity i.e. 2000 kl/hr, failed on account of split muff coupling Allen bolts failure. So it was decided to carry out preventative maintenance of booster pump No 04. The maintenance of the booster pump No 04 was started from 21.01.2014.**

4) Upon dismantling of the pump following are the observations:-

- One crack was observed in the 2nd stage case wear ring of the pump.
- Sleeve seal which is integral part of the Mechanical seal was found seized with shaft.

5) Case wear rings are stationary components and are designed to prevent wear & tear between the impeller and pump casing. The crack in the 2nd stage case wear ring might have occurred due to material failure as overheating or rubbing marks were not observed on its surface. Fig-1 attached as Annexure-I.

6) **The mechanical seal sleeve was found seized with the shaft and that might have occurred due to overheating. The overheating may have been resulted because of foreign particles between shaft and sleeve.**

7) Precaution to be taken during restoration and future action plan:-

- Alignment of pump and vibration monitoring.
- Replacement of muff coupling bolt in line with BP3.
- Inspection of Y-type strainer
- Failure analysis of case wear ring may be done for in-depth analysis of the failure.

Submitted please.

28/1/14
(Amit Kumar)
DMO, Gauridad

SMNM 30/1/14

CMNM

Out of 5 new boosters at Vadinar, 4 boosters (BP1, BP2, and BP3 & BP4) have recently undergone major overhauling due to problems observed during operation. BP1, BP2 & BP3 were not developing discharge pressure and BP4 had tripped in motor short circuit. On dismantling the pumps of BP1, BP2 & BP3, various components as listed below were found damaged.

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NOTE

Booster Pump No 01:- (29.15 running hours)

1. Out of the four split muff couplings, only one split coupling was found intact.
2. Two Allen bolts of 2nd and 4th split muff couplings were found sheared.
3. All the eight bolts in one split muff coupling were found sheared.
4. One discharge bowl bush was found with marks, which was replaced with new one.
5. One semi-circular split ring of split muff coupling was sheared.

Booster Pump No 02:- (1212.07 running hours)

1. All Allen bolts of 1st split muff couplings were found sheared.
2. One Allen bolt of 3rd split muff coupling was found sheared.
3. Two Allen bolts of 4th Split muff coupling was found sheared.
4. Two nos.Holes of 1st stage impeller retaining plate became oval.
5. Three nos. Holes of 2nd stage impeller retaining plate became oval.
6. Mechanical seal was found seized.

Booster Pump No 03(871.23 running hours)

1. **One Allen bolt of 3rd split muff couplings was found sheared**
2. All the Allen bolts of 4th split muff coupling were found sheared.
3. No washers were found beneath Allen bolts of retaining plate.

In spite of the fact that in BP4 there was no abnormality observed in the pump during operation, however, in order to check the internal components and to replace the coupling bolts with similar bolts , as have been used in BP1, BP2 & BP3, it was decided to dismantle BP4 pump also. On dismantling the pump of BP4, one case wear ring was found cracked and the mechanical seal sleeve was found seized with shaft. The cumulative running hours of BP4 is 1494.16

The probable reasons for the failure of components of BP4 pumps are as follows:

Case wear ring: The crack in the case wear ring might be due to material problem. The material of case wear as per manufacturer standard is ASTM A 743 GR CA40. It is cast steel. To identify the reason of failure, testing of material composition of the cracked case wear ring is being planned through any reputed laboratory.

Mechanical seal sleeve: The seal sleeve was found seized with the shaft probably due to overheating caused by foreign particle between shaft and sleeve.

Since, similar problems as encountered in above four boosters cannot be overruled in BP5 also, it is advisable to check the condition of the pump of BP5 also. The dismantling of the pump for inspection has been planned from 3rd Feb 2014.

Submitted please.

DGM (O)

3.2.2014

एस. डी. कामेबाड़
डी जी एम (ओ)

(P.K.SARKAR) 30/01/2014
CMNM

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- Out of five Booster at Vadinar 3 No having capacity of 2000 kl/hr, in all the 3 Booster problem of split muff coupling bolts shearing was observed it is clear that M/s BPCL has not designed the coupling properly. This is need to be taken up with M/s BPCL.

- Further in small capacity pump i.e BP-4 flash in winding was observed, presently motor is at Mumbai in M/s Ansaldo's authorized workshop for rewinding. Further in BP-4, pump crack was observed in case wear ring & sleeve seal of mechanical seal was found seized with shaft.

- In view of above, it is suggested that matter need to be take up with M/s BPCL by project HO, considering all design aspect & frequent breakdown, as above.

3.2.2014

एस. डी. कामेबाड़
डी जी एम (ओ)

GM, WRPL

The subject matter related to Boosters at Vadinar was also mentioned during PMC meetings. In view of repeated failures in newly installed boosters, it is proposed that matter may please be taken up with M/s BPCL by Project deptt PLHO.
Submit for consideration pl.

13/2/2014

ED WR

forwarded for kind perusal & further need ful pl.

13/02/2014

ED (PS) - I/c

DGM (MNT) on leave

का.नि. (प.क्षे.)
दि. 13/2/14

CC: ED (C) For kind information

The matter was discussed both D(PL) & CMD BPCL during D(PL)'s visit to Allahabad. BPCL have agreed to rectify all problem by March. Please follow up.

19.2.14 GM(MCS) 20/2/14